A Flood Prediction Method Integrating MSSU and VGDAG-STNN Under Limited Data

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Abstract

- **Flood warning** in small basins is hindered by **sparse data** and **static models**.
- We propose an **end-to-end system** integrating low-cost **MSSU** (Mobile Spherical Sensing Unit) and **VGDAG-STNN** (Spatiotemporal Neural Network Based on Velocity-Guided Dynamic Acyclic Graph).
- Proposed **VGDAG** (Velocity-Guided Dynamic Acyclic Graph) updates **graph topology** in real time to capture **upstream-downstream flow correlations**. Trained with only 33 simulation samples, our model achieves a **60-min forecast with R^2=0.999**, cutting **MSE** by 88.2% against **LSTM**.
- Field tests verify MSSU stable data acquisition. This physics-data hybrid framework provides low-cost flood warning for data-sparse watersheds.

1. Introduction

- Global floods cause $180B annual losses; China's flood damage exceeds 80 billion RMB each year.
- Small & medium river basins lack dense fixed hydrological monitoring equipment.
- Goal: Build sensing-model integrated system to get accurate 60-min forecast with limited training data.

2. Literature Review

- Traditional hydraulic models & static GCN require abundant data and cannot adapt dynamic water flow.
- Existing dynamic graph algorithms (traffic prediction) lack hydrodynamic physical constraints.
- Research Gap: No low-cost sensing hardware + physics-guided dynamic graph for small-sample flood forecasting.

Our solution: MSSU+ VGDAG-STNN

3. Methodology

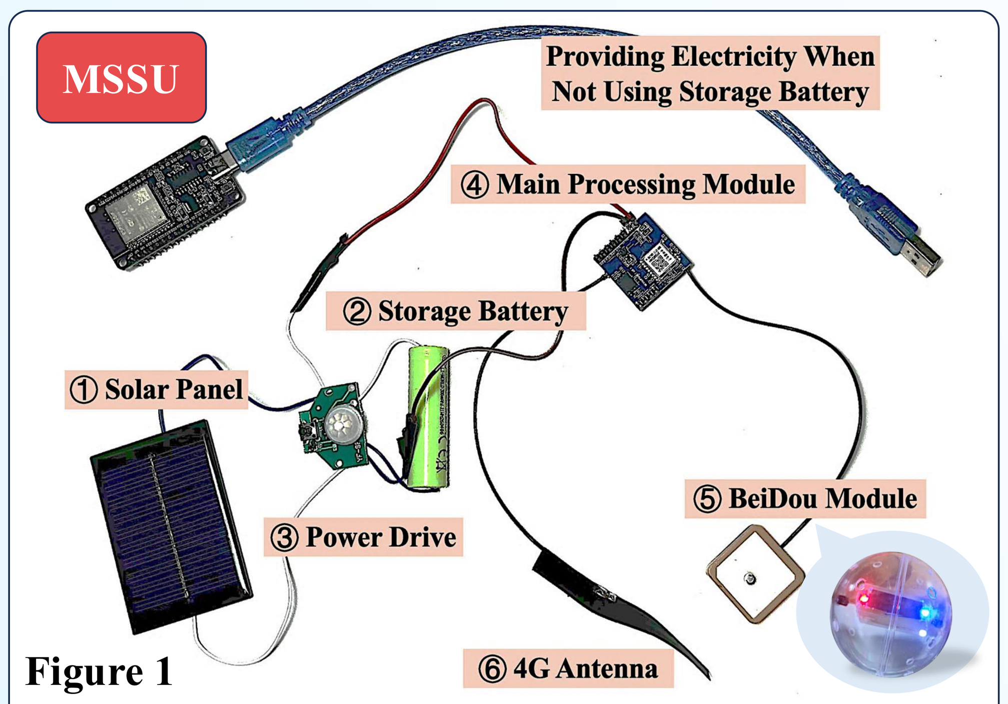

Figure 1
Self-Developed MSSU Prototype of the Proposed System

★ As shown in Figure 1, **MSSU** is a UAV-launched 7D flow probe with Beidou+4G. Field test: valid packet rate in urban 100%, mountain area 62.2%.

Energy Flow: ① Solar Panel (generates electricity) → ② Storage Battery (stores) → ③ Power Drive (regulates voltage + supplies stable power) → ④ Main Processing Module.

Data Flow: ④ Main Processing Module (wakes up + sends commands) → ⑤ BeiDou Module (measures altitude + return altitude to main module) ④ Main Processing Module (packages data) → ⑥ 4G Antenna (uploads to cloud server)

6. References

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3. Methodology

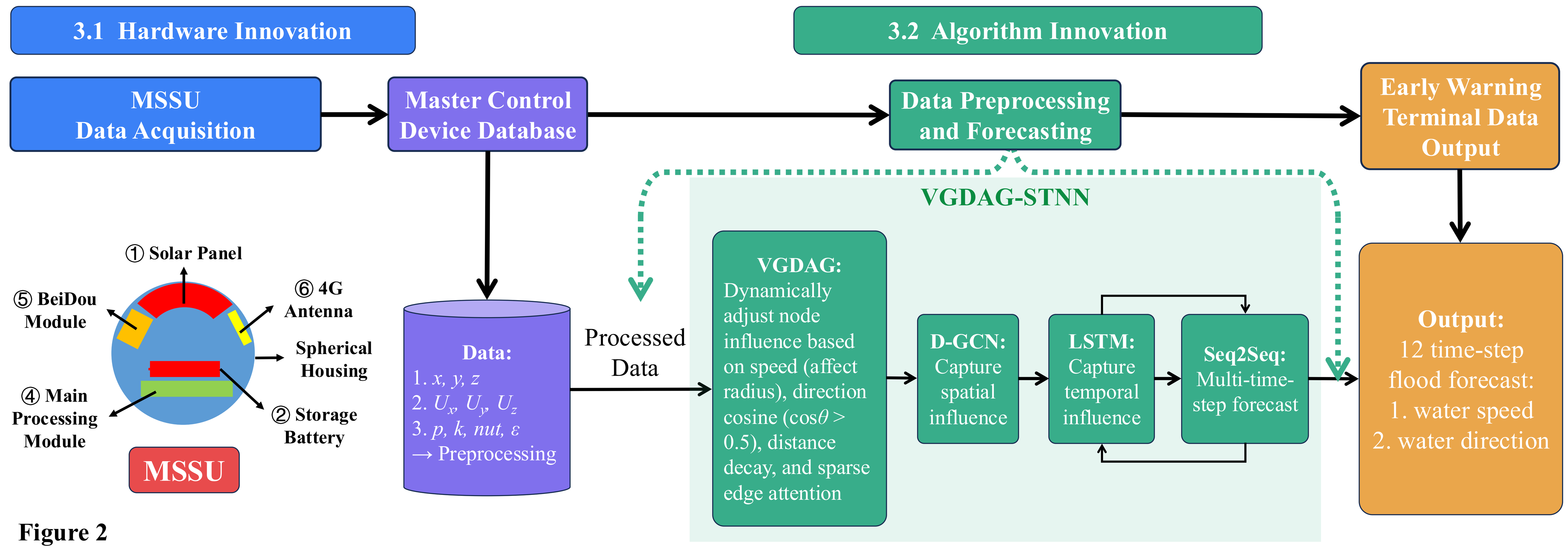

Figure 2
Overall Architecture of the Proposed System

According to Figure 2, the framework follows a sequential pipeline: **MSSU Data Collection → Preprocessing → VGDAG Construction → D-GCN-LSTM-Attention → 12-step 60-min Velocity Prediction**

Input: 32 historical time steps | **Output:** 12 forecast steps | **Dataset:** 33 sliding-window CFD samples

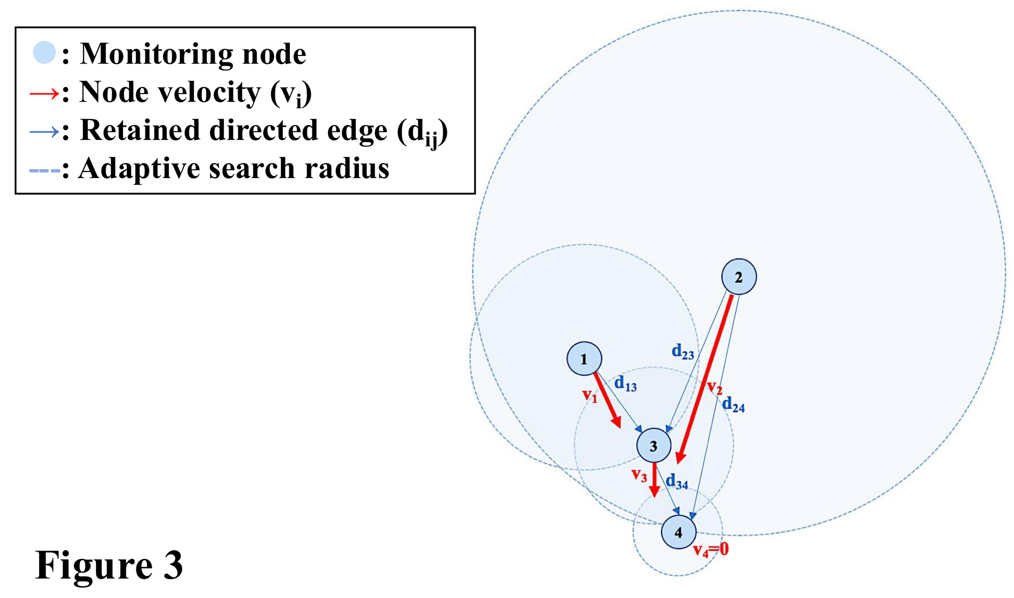

Figure 3
Dynamic Topology Construction of VGDAG

3.3 VGDAG Construction

VGDAG constructs **dynamic edges** in **real-time** based on **instantaneous flow velocity** and **direction.**

As shown in Figure 3, the **dynamic graph** is constructed via the following steps:

① Calculating **node velocity**
② Defining an **adaptive search radius**
③ **Restricting** connections to **downstream** directions via **neighborhood screening**
④ Assigning **distance-decayed edge attributes** for graph convolution

The related formulas are listed below:

Node Velocity

$$v_i=\sqrt{Ux_i^2+Uy_i^2+Uz_i^2}$$

Adaptive Search Radius

$$r_i=\text{clip}(r_0+\alpha\cdot v_i, r_{\min}, r_{\max})$$

Neighborhood Screening

$$cos\theta=\frac{\mathbf{U}\cdot\mathbf{d}}{\|\mathbf{U}\|\|\mathbf{d}\|}>0.5$$

Edge Attributes

- Euclidean distance d
- Directional cosine $\cos\theta$
- Source node velocity v_s

4. Results

Table 1
Overall Prediction Performance Comparison

Model	*MSE*	*RMSE*	*MAE*	R^2
LSTM	0.00495	0.0703	0.0424	0.995
GCN-LSTM	0.00384	0.0619	0.0352	0.996
VGDAG-STNN	0.000584	0.0242	0.0182	0.999

Table 1 shows that **VGDAG-STNN outperforms across all metrics**, **reducing MSE by 88.2%** (vs. LSTM) **and 84.8%** (vs. GCN-LSTM), with an **R^2 of 0.999.**

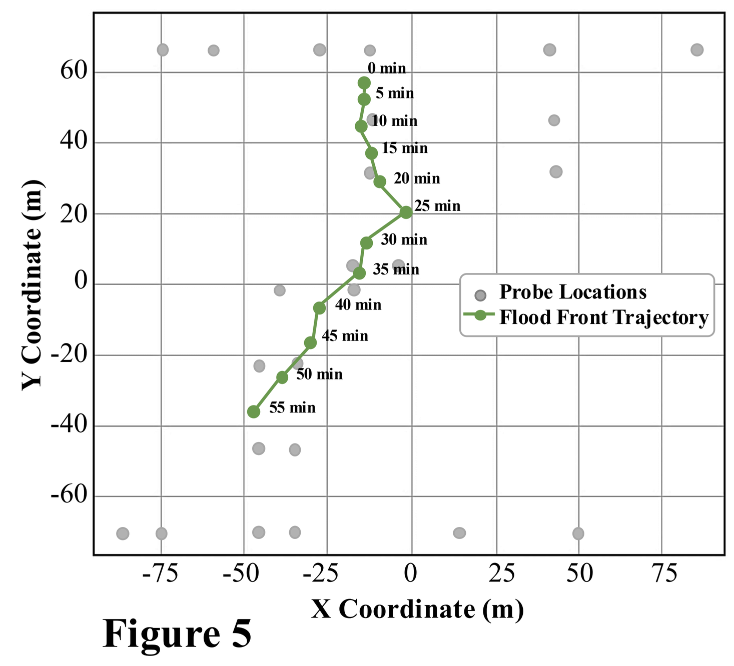

Figure 5
Real Water Flow Trajectory

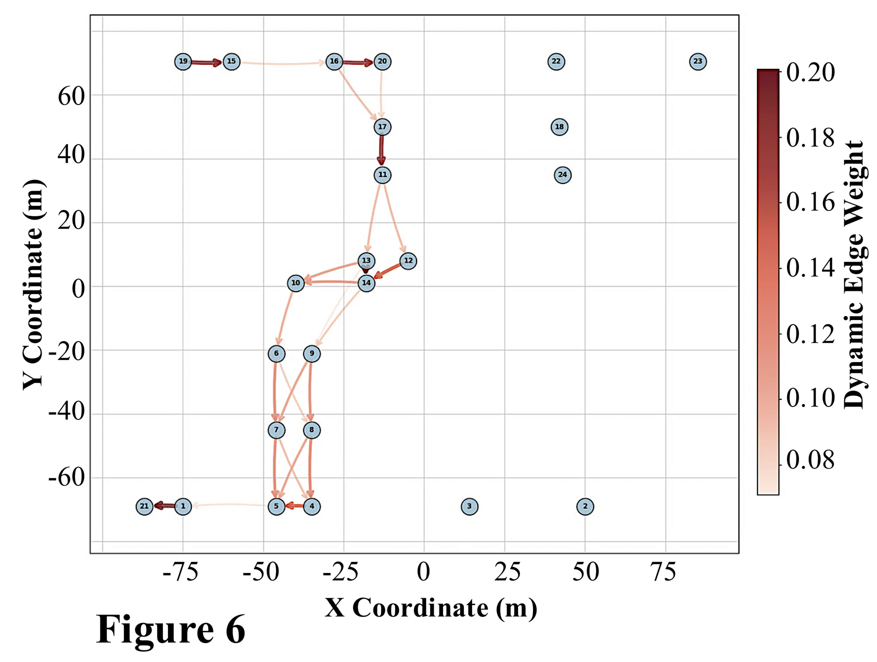

Figure 6
Dynamic Graph Constructed by VGDAG

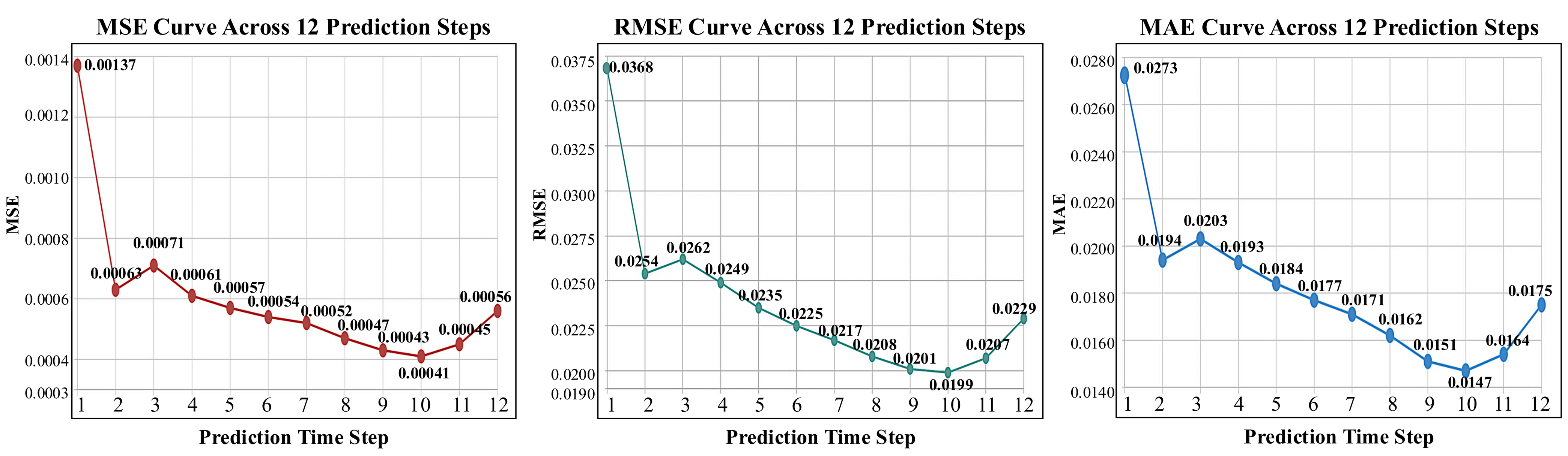

Figure 4
Long-term Forecast Error Stability

Figure 4 verifies long-term stability. Errors **start slightly high** at step 1, decrease to **an optimal minimum** by step 10, and only **rise marginally** at 11-12, proving **long-term stability without error accumulation or drift.**

Comparing **dynamic graphs** (Figure 6) constructed by **VGDAG** with the **real trajectories** (Figure 5), the edge connections **evolve over time and align closely with the physical migration path**, confirming that **VGDAG captures realistic upstream-downstream correlations.**

5. Discussion & Future Work

Key Findings

- Field-verified **MSSU** solves small-basin monitoring gaps.
- Proposed **VGDAG** models dynamic river upstream-downstream relations.
- High-precision prediction with 33 samples.

Limitations

- Validated only on CFD simulation, lack real flood field data.
- MSSU signal weakens in mountain regions.
- VGDAG parameters need manual adjustment for different basins.

Future Work

- Integrate rainfall, terrain and remote sensing data.
- Improve MSSU communication for weak-signal mountains.
- Carry out real river field verification.